SAPHIR: Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous

Rathod Vaibhav Vishwas Ram
Department of Computer Science and Engineering, R. V. College of Engineering, Bengaluru, India
github.com/vaibhav-rm/SAPHIR

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Abstract

Voice AI agents that encounter each other on a call still converse the wasteful way — speech → text → LLM → speech on both sides — even though both endpoints are machines. The viral Gibberlink demonstration (February 2025) showed two agents switching to data-over-sound tones instead, but its switch is unauthenticated, unnegotiated, unstructured, and opaque to human operators. Meanwhile the emerging interoperability stack (MCP, A2A/ACP, ANP, InterSAGE) uniformly assumes IP connectivity over HTTPS, leaving voice-only calls, PSTN trunks, RF-restricted and offline settings without a pairing story. This paper presents SAPHIR, a minimal authenticated pairing protocol for two heterogeneous agents that share nothing but a bare audio channel: a three-phase handshake (HELLO → WELCOME → KEY1 → KEY2) with ephemeral X25519 keys, compact CBOR task encoding, ChaCha20-Poly1305 authenticated messaging, capability advertisements and session resumption in the handshake, an optional stop-and-wait ARQ layer for lossy acoustics, and a human-readable sidecar audit log. An open prototype (∼2.3k lines of Python including tests and harnesses) demonstrates a hotel-booking dialogue end to end: 117–139 B task frames across two domains, blocked spoof/replay/downgrade attacks, 100% pairing success on clean channels, 95–100% under bursty VoIP-like loss where raw frames score 0%, and immediate cost breakeven that widens to ∼65× at 20 turns. We state precisely what is measured, what is modeled, and what remains — over-air validation, PKI-backed identity, and group sessions — and close with open research questions.

1 Introduction

In February 2025 a hackathon project called Gibberlink [1] demonstrated two ElevenLabs voice agents on a phone call discovering both sides are AI and switching from English to `ggwave` [2] data-over-sound tones. The video passed 15 M views and the intuition is sound: rendering speech for a machine listener wastes two neural speech passes and thousands of tokens per turn. The implementation, however, is a prompt ("switch if the other side is AI") plus raw tones. Four gaps follow directly: (1) the switch is *unauthenticated* — any speaker can emit beeps; (2) the format is *unnegotiated* — both sides must hardcode the same profile; (3) the content is *unstructured English* — tones carry sentences, not intents; (4) the exchange is *opaque to humans* — operators cannot read what was agreed. A recorded beep replays; a crafted beep injects; a legacy peer is silently beeped at.

In parallel, 2024–26 produced the first serious agent-interoperability stack: Anthropic's MCP for agent-to-tool access, Google's A2A (now Linux Foundation) for task delegation via Agent Cards, IBM/Cisco ACP variants for performative messaging, ANP for decentralized discovery with W3C

DIDs, and InterSAGE as the identity/trust substrate [3, 4]. Security analysis is maturing alongside: A2ASecBench [5] and ASB systematize protocol-logic attacks that bypass default safeguards, and communication-graph analysis [7] shows even encrypted transports leak who-talks-to-whom. Yet every protocol in this stack assumes URL-addressable agents over HTTP(S). IETF framework drafts [9] list discovery, negotiation, and fallback among open requirements. None answers: *what if the only shared channel is sound?* Near-term agent telephony (AI receptionists, AI assistants calling businesses), PSTN trunks that destroy ultrasound, RF-restricted facilities, and offline/disaster settings all produce voice-only encounters between heterogeneous agents.

Contribution. We argue acoustic pairing deserves first-class protocol status and present SAPHIR (Secure Acoustic Pairing for Heterogeneous Inter-agent Rendezvous): detect, negotiate, encrypt, collaborate in structured intents, and account for every exchange in a human-readable sidecar. We contribute (i) the protocol (handshake, compact encoding, capability discovery, resumption, AEAD messaging, optional ARQ); (ii) an explicit threat model with verified blocks; (iii) an open prototype with seven evaluation harnesses; and (iv) a disciplined measured-vs-modeled accounting that states exactly what is (and is not) claimed. All artifacts are public to enable red-teaming.

2 Background and related work

Gibberlink and ggwave. Gibberlink [1] contributes the core observation (speech is wasteful machine-to-machine) and a demo, not a protocol: no framing beyond ggwave frames, no handshake, no security. `ggwave` [2] is a data-over-sound modem (FSK + Reed-Solomon, tens of bytes/sec in robust audible profiles) in the lineage of dial-up modems; SAPHIR treats it as a byte pipe and adds everything above it.

The HTTP interop stack. The four-protocol survey [3] (MCP/ACP/A2A/ANP) proposes a phased roadmap from tool access to decentralized agent marketplaces — all over IP. InterSAGE [4] supplies the missing trust substrate (persistent identity, capability-aware discovery, attenuated delegation, audit trails) and explicitly complements, rather than replaces, transport protocols. SAPHIR's claim is narrow and compatible: it is the pairing layer for the one transport the stack omits, and its ephemeral-DID handshake is designed to upgrade to InterSAGE-style credentials where IP exists (hybrid mode).

Attack literature. A2ASecBench [5] taxonomizes supply-chain and protocol-logic attacks across A2A stages; ASB [6] benchmarks prompt-injection, memory poisoning, and backdoors across agent backbones; both find high attack success against default configurations. Our threat model imports their methodology (named attacker capabilities, verified blocks) at the acoustic layer, where the cheapest attacks — replay and injection — need only a speaker.

Semantics and metadata. Two recent critiques shape our design: well-formed JSON does not imply shared meaning [8], motivating our ontology-tagged intent slots over raw English; and encrypted content still exposes the communication graph, whose action-coupled metadata threatens workflow integrity [7], motivating our sidecar's selective disclosure as future work rather than a solved claim.

Deployment context. Industry surveys report the trust gap blocking agentic-AI production (85% exploring, 5% in production) [10]; IETF drafts [9] formalize the missing negotiation/fallback machinery. SAPHIR targets exactly that intersection for voice-only encounters. Table 1 summarizes the positioning: every prior row standardizes content while assuming IP transport; SAPHIR is the pairing layer for the one transport the stack omits.

Table 1: Acoustic pairing against the 2025–26 interop stack. SAPHIR complements rather than replaces these protocols (cf. InterSAGE's stance toward transports).

System	Channel	Auth	Negotiation	Audio fallback
Gibberlink	audio only	none (prompt)	none	itself, insecure
MCP	HTTPS (tool)	yes	tools	none
A2A	HTTPS (agent)	yes	Agent Cards	none
ANP	HTTPS + DID	yes	DIDs	none
InterSAGE	trust substrate	yes	credentials	none
SAPHIR	**audio / hybrid**	**ephemeral keys**	**codec select**	**native**

3 Problem and threat model

Two heterogeneous voice agents share a low-bandwidth (∼30 Bps reference), possibly lossy, possibly adversary-observable audio channel, plus optionally an IP side channel (hybrid mode). Goals: pair within seconds, agree on codecs and a task ontology, communicate confidentially with integrity, degrade explicitly (never silently) with legacy peers, and leave a human-inspectable record. Attacker capabilities: observe, record, replay, and emit arbitrary audio; impersonate without keys; force or suppress the switch. Out of scope in this version: attackers holding stolen session keys (needs PKI rotation), traffic-analysis resistance, and denial-of-service on the acoustic medium.

4 The SAPHIR protocol

Framing. Length-prefixed CBOR envelopes, `u32be len` ‖ `CBOR({v,type,seq,sender,nonce,payload})`, cap 200 B (enforced). **Handshake.** HELLO (ephemeral `did:key`, integer codec/mode codebooks, plus capability advertisements: the initiator's `need` and the sender's `have` bitmask over registered intents with a 10 B bloom filter for open vocabularies) → WELCOME (selected codec plus an `exact/maybe/no` capability verdict; peers offering no SAPHIR codec are flagged and kept in English) → KEY1/KEY2 (32 B X25519 ephemeral pubkeys each way; HKDF-SHA256 derives a 32 B ChaCha20-Poly1305 session key; forward secrecy per pairing). **Resumption.** After a full pairing both sides store a ticket under the peer's stable deployment identity; return visits send {`resume, cr`} and complete on {`resumed, sr`} with keys = HKDF(psk, cr‖sr), fresh every visit, skipping KEY1/KEY2 (273 B vs 451 B measured). The responder rests in RESUMING until the first successful AEAD open, so resumption authenticates implicitly; tickets expire by time and uses, and misses fall back silently. **Messaging.** TASK/RESULT carry `ChaCha(cbor({intent,slots,ontology,ts}))` with the sender as AAD; replays are rejected by sequence plus (nonce, ciphertext) sets. **Compact encoding.** Hotel-ontology shorts (`hotel.book`→`hb`, `date`→`d`, . . .) cut task frames 182/202 B → 117–139 B across hotel and taxi domains (non-default ontology tagged, value shorts); legacy full-form still decodes. **ARQ (optional).** Stop-and-wait layer: ≤110 B CRC32 chunks (each fits one audible frame) with ACKs and bounded retries, tick-driven and deterministic. **Audit.** Every frame is appended to a JSONL sidecar with a human summary and decrypted view. Figure 1 summarizes the exchange.

5 Implementation

The prototype (`github.com/vaibhav-rm/SAPHIR`, MIT) is ∼2.3k lines of Python: `protocol` (envelopes), `session` (handshake state machine), `crypto` (X25519/HKDF/ChaCha), `codec` (compact

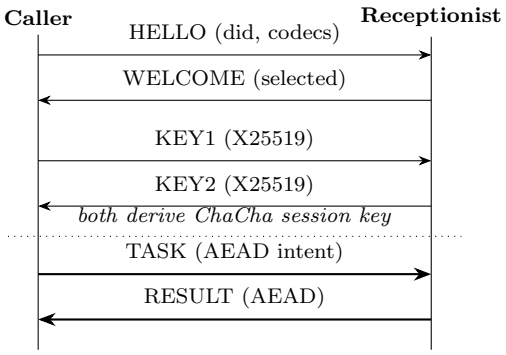


Figure 1: SAPHIR pairing. Negotiation is plaintext; tasks are encrypted; every frame is sidecar-logged.

maps), `caps` (capability bitmask + bloom), `resume` (tickets), `channel` (in-memory, lossy, burst-error, ARQ link), `agents` (rule-based hotel and taxi tasks), `audit`, `metrics`. Ten harnesses (`demo`, `bench`, `attacks`, `robustness`, `reliability`, `codec_profiles`, `amortization`, `ggwave_frames`, `decoder_fragility`, `mitm`) plus 28 pytest tests cover functionality, attacks, and all sweeps. No audio hardware, API keys, or network are required; channels are in-process byte pipes with documented seeds.

6 Evaluation

Functionality. Full pairing plus a two-intent booking dialogue runs end to end; measured sizes: handshake 463 B one-time, HELLO ∼115 B, WELCOME ∼128 B, KEY ∼107–113 B, all four TASK frames 117–139 B compact (hotel and taxi domains), RESULT 126–139 B — every frame fits one ggwave transmission. Local compute is negligible next to airtime (full handshake 0.22 ms, task seal/open ∼10–20 μs measured).

Attacks (Table 2). A keyless attacker is rejected (`InvalidTag`); exact replays are rejected (sequence reuse); legacy peers are kept in English with an explicit flag. All three ship as failing-closed checks in `evals/attacks.py`.

Table 2: Threat outcomes: raw tones vs. SAPHIR (prototype-verified).

Attack	Raw tones	SAPHIR
Tone spoof / injection	succeeds	blocked (AEAD tag)
Replay of valid frame	succeeds	blocked (seq + nonce set)
Forced downgrade	silent	flagged, stays in English
Eavesdropping	plaintext	ciphertext
Impostor responder (own keys)	pairs + reads tasks, zero signal	needs PKI (demoed)
Stolen session keys	n/a	out of scope (needs PKI)
Who-talks-to-whom	n/a	out of scope

Raw-channel sweep (50 trials/cell, no retransmission): 100% clean, 94% at 1% loss, 44% at 10% loss; corruption cliff (36% at 0.1%, 0% at 0.5%) since AEAD must reject forgeries — integrity over availability, by design. *ARQ sweep (20 trials/cell, Table 3):* the same grid converts the cliff into retry cost: 100%/95% where raw scored 0%, at ∼1.6× wire bytes; residual 10%-loss failures are bounded-budget retry exhaustion.

Table 3: ARQ vs. raw success rate per grid cell (drop, corruption).

Cell	Raw	ARQ	Median retries (ARQ)
(0.00, 0.000)	100%	100%	0
(0.05, 0.000)	70%	100%	1
(0.00, 0.005)	0%	100%	8
(0.05, 0.005)	0%	95%	12
(0.10, 0.000)	44%	85%	3

Capability discovery: three-valued matching over ∼13 B of HELLO (exact bit, bloom `maybe` at 0.25% measured FPR, fast `no` with a plaintext decline) — preliminary evidence that capability semantics fit the acoustic budget; schema-driven open vocabularies remain future work. *Resumption:* return visits pair in 2 frames for 273 B (∼40% saved), with expiry-forced fallback and keyless-attacker rejection verified. *Codec/noise profiles (byte-level burst models, 20 trials):* clean 100%, cafe-noise 95%, bursty VoIP-like 95%, harsh narrowband 80% — bounding the design's operating envelope; waveform/over-air measurement is explicitly future work. *Amortization (Fig. 2):* the 463 B handshake amortizes over turns (mean 246 B wire per pair); modeled neural-cost breakeven is immediate and ∼65× at 20 turns. Wall-clock is deliberately not claimed: narrowband audio moves fewer bytes/sec than speech, so SAPHIR trades time for near-zero neural compute; wire bytes stay above raw-text-over-tones (structure plus auth tags), shown, not hidden.

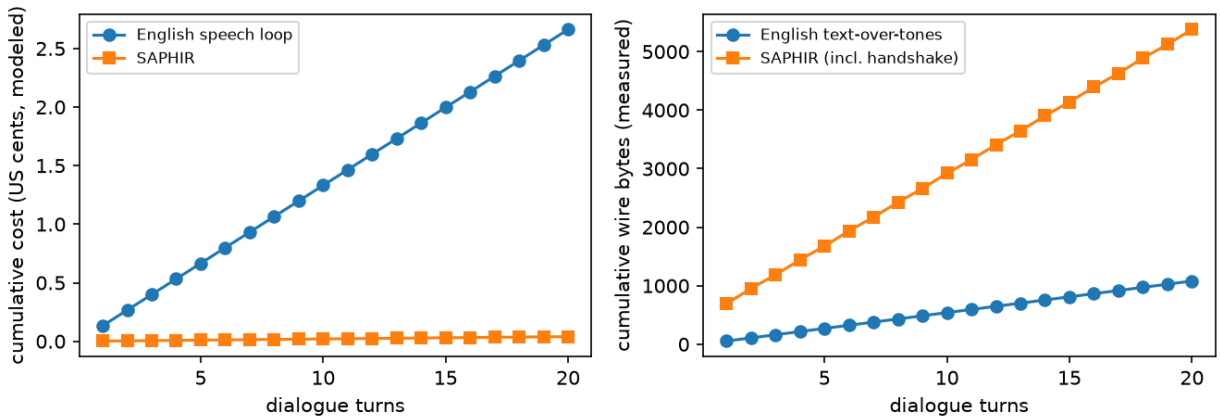


Figure 2: Cumulative modeled cost (left) and measured wire bytes (right) vs. dialogue turns. Cost wins immediately; bytes do not — by design and on the record.

7 Limitations and open questions

(1) No room-acoustics validation yet — loss models are byte-level, though all framing now round-trips through the real ggwave implementation in-process and a browser demo drives genuine over-air exchange; Opus/G.711 traces and measured rooms are the next experiment. Two learned lessons belong here: the ggwave implementation shows content-dependent first-try decode misses, so retries must re-seal with fresh randomness rather than retransmit identical bytes; and single-shot in-process decoding is fragile to any perturbation (both builds), which is why loss is modeled at the byte level and retry is load-bearing. (2) Identity is ephemeral, not PKI-backed; upgrading the

HELLO to verifiable credentials (where IP exists) is designed but unbuilt. Hand dictionaries are demo-scale; schema-driven dictionaries negotiated in HELLO are the scalable compression story and remain future work. (3) Two-party only; MLS-style groups are future work. (4) Auditing stops at readability; selective disclosure without workflow-graph leakage [7] is open. We invite red-teamers: the artifact reproduces every table from one command each (see `docs/REPRODUCIBILITY.md`).

8 Conclusion

Gibberlink asked the right question — why speak to machines in speech? — and gave a demo answer. SAPHIR gives a protocol answer for the channels the interop stack forgot: negotiate briefly, encrypt cheaply, speak in structure, and keep every beep readable. The evidence here is preliminary but exact about its boundaries; the operating envelope, the cost curves, and the open challenges are published alongside the code so the community can push past them.

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